

Course: Object Oriented Programming

Lab#5: Array Manipulations

Task 1:

You are a software engineer developing a mobile app for a book-reading service. A feature requires reversing an array that represents chapter numbers, but the chapters should be reversed only if the array length is even. If the array length is odd, reverse only the odd-indexed elements and leave even-indexed elements unchanged.

Requirements:

1. Take an array of integers as input from the user (representing chapter numbers).
2. If the array length is even, reverse all elements.
3. If the array length is odd, reverse only the elements at odd indices (e.g., indices 1, 3, 5, etc.).
4. Print the original and modified arrays.

Sample Output:

```
Enter the number of chapters: 5
Enter chapter numbers: 10 20 30 40 50
Original Array: [10, 20, 30, 40, 50]
Reversed Array (odd indices): [10, 40, 30, 20, 50]

Enter the number of chapters: 6
Enter chapter numbers: 5 10 15 20 25 30
Original Array: [5, 10, 15, 20, 25, 30]
Reversed Array: [30, 25, 20, 15, 10, 5]
```

Task 2:

You are creating a leaderboard for a local cricket tournament. The leaderboard lists player scores. You need to find the second-highest unique score but must handle cases where all players scored the same.

Requirements:

1. Take an array of integers as input, representing player scores.
2. Find the second-highest unique score.
3. If there are not enough unique scores, display "Not enough unique scores."

Sample Output:

```
Enter the number of players: 6
Enter the scores: 100 90 80 90 100 70
Second highest score: 90

Enter the number of players: 4
Enter the scores: 50 50 50 50
Not enough unique scores.
```

Task 3:

You are tasked with building a weather prediction app. A feature requires rotating the daily temperatures array to the right by a specified number of days, but only rotate the temperatures that are above a given threshold.

Requirements:

1. Take the array size, elements (daily temperatures), rotation count, and a threshold temperature as input.
2. Rotate only the elements greater than the threshold while keeping others in their original positions.
3. Print the original and rotated arrays.

Sample Output:

```
Enter the number of days: 7
Enter the temperatures: 30 40 25 45 50 20 35
Enter the rotation count: 2
Enter the threshold temperature: 30
Original Array: [30, 40, 25, 45, 50, 20, 35]
Rotated Array: [30, 50, 25, 35, 40, 20, 45]
```

Task 4:

You are working on an inventory system for a warehouse. Two departments provide lists of product quantities. Merge the lists and sort the resulting array in ascending order. However, if a product's quantity appears in both lists, its value must be excluded from the final array.

Requirements:

1. Take two arrays of integers as input.
2. Merge the arrays, excluding any duplicate values that appear in both arrays.
3. Sort the final array in ascending order and display it.

Sample Output:

```
Enter the size of the first array: 5
Enter the elements of the first array: 10 20 30 40 50
Enter the size of the second array: 5
Enter the elements of the second array: 30 40 60 70 80
Merged and Sorted Array (with duplicates removed): [10, 20, 50, 60, 70, 80]
```

Task 5:

You are designing a payment reconciliation system. A list of payment amounts is given, and you need to find all pairs of amounts that sum up to a target value. However, any pair where one amount is greater than half of the target should be excluded.

Requirements:

1. Take an array of integers (payment amounts) and a target sum as input.
2. Find all unique pairs of elements whose sum equals the target.
3. Exclude pairs where one amount is greater than half of the target.

4. Print the unique pairs that meet the criteria.

Sample Output:

```
Enter the number of payments: 6
Enter the payment amounts: 20 30 40 50 60 70
Enter the target sum: 90
Valid Pairs: (20, 70), (30, 60), (40, 50)

Enter the number of payments: 5
Enter the payment amounts: 25 25 30 45 60
Enter the target sum: 60
Valid Pairs: (25, 30)
```